

EXPERIENCE

SYDNEY MOTORSPORTS

Lead Aerodynamicist

Sydney, Australia

2022.07 - Current

DOVETAIL ELECTRIC AVIATION

Computational Fluid Dynamics Intern

Sydney, Australia

2023.09 - 2024.01

TECHNICAL SKILLS

Composite Manufacturing

Part of the manufacturing of a aerodynamics package for Sydney Motorsports through both wet layup and prepreg. The molds were made from high density foam shaped by a variety of equipment such as CNC hot wire cutting, CNC routing and hand sculpting.

Computer Aided Design

Modified parts of the aerodynamic package for Sydney Motorsports as well as creating the baseline simulation geometry for Dovetail Electric Aviation. All CAD projects were made to be fully parameterized to allow for simple iterations of the design. Designed various iterations of gearboxes for robotics applications in Autodesk Inventor.

Numerical Simulation

Wrote a semi-automated workflow for testing aerodynamic component designs over a range of angles for sensitivity studies. Created a simple CFD model of a aircraft engine assembly including mesh analysis study including simulating a radiator in order to study the cooling effects. Conducted a thermal study in both fluent and Steady State Thermal of how much the heat output of a human and two computers would heat up a room.

Programming

Capable of writing analysis code in MATLAB and Python. Created a blade element study in MATLAB to study the performance of various propellers' performance under different flight conditions. Wrote custom FEM code and compared it with the commercial Steady State Thermal package.

TECHNICAL PROJECTS

Semi-Automated CFD Workflow

Worked with and improved the robustness of the previous semi-automated CFD workflow in Sydney Motorsports through experimenting with the mesh settings. This allows more designs to be iterated through the workflow.

Aerodynamic Package for FSAE Car

Involved planning the manufacturing, mounting and validation of the second in house aerodynamics package for Sydney Motorsports. Multiple relatively simple and low cost methods are planned such as utilizing foam core and carbon honeycomb sandwich structures to increase the ease of manufacturing and strength of the components produced. Mounting solutions are improved to make track-side assembly and maintenance significantly faster.

Custom Cloud Storage Server

Created a custom server from e-waste and second hand parts for self-hosted cloud storage solutions similar to OneDrive with a RAIDZ5 configuration. The system also acts as a media server and NAS for additional storage, and can be access remotely through VPN.

UNIX OS

Dual booted various distros of UNIX such as Ubuntu and Arch alongside Windows on my PC out of interest for light-weight OS. This allowed me to be comfortable with TUI commands as well as familiarity with VIM.

Robotic Actuator

Designed Cycloidal gearboxes with various reduction ratios for general gear reduction as well as robotic purposes, primarily targeted at stepping down the RPM of BLDC motors so high torque can be generated at increased accuracy.

Topology Optimization

Optimized the design of a rib and spar system for the front wing of the FSAE car with different materials to find a balance between performance, cost, manufacturing and time constraints.

TRANSFERABLE SKILLS

Problem Solving

Rapid response to critical failures during composite manufacturing of aerodynamic surfaces to ensure the part is made and minimal waste is produced. Resolving design challenges that are presented while minimizing weight and cost.

Communication

Technical details from other departments within the team need to be communicated clearly in order to ensure the design produced is realistic and will work together.

Time Management

Ensuring that personal design deadlines are met for the Formula SAE team as well as Dovetail Electric Aviation while studying full time.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics) - Credit WAM

Expected Graduation: 2025

2021 - Current